EDS Practical :05

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5.1.1 Stacked plot C

od e:

```
import matplotlib.pyplot as plt import

pandas as pd

# Data for Months and Temperature for three cities data = {

    'Month': ['January', 'February', 'March', 'April', 'May',
  'June', 'July', 'August', 'September', 'October', 'November',
  'D e c e mbe r '],

    'City_A_Temperature': [5, 7, 10, 13, 17, 20, 22, 21, 18, 12,
  8, 6],

    'City_B_Temperature': [2, 3, 5, 6, 10, 14, 16, 17, 12, 9, 5,
  3],
        'City_C_Temperature': [3, 4, 6, 8, 9, 12, 15, 14, 10, 7, 4, 2]
  }

  # Write your code... df =

  pd.DataFrame(data)

  plt.stackplot(df['Month'],df['City_A_Temperature'],df['City_

  B_Temperature'],df['City_C_Temperature'])

  plt.xlabel('Month') plt.ylabel('Temperature')
```

plt.title('TemperatureVariation') plt.legend(loc='upper left')

plt.show()

5.2.1 Titanic Dataset C

od e:

import pandas as pd import

matplotlib.pyplot as plt

Load the Titanic dataset from the CSV file df =

pd.read_csv('titanic.csv') # Set up the figure for

5 subplots

fig, axes = plt.subplots(3, 2, figsize=(12, 12))

write the code..

```
##plot 1:count of passengers by class axes[0,
0].bar(d f['Pc l ass'].v al ue _ c o unt s( ).i nd e x , d f['Pc l ass'].v al u
counts(), color='skyblue')
axes[0, 0].set_title("Passenger Class Distribution")
axes[0,0].set_xlabel("Pclass") axes[0,0].set_ylabel("Count")
axe s[0, 1].pi e (d f['Ge nd e r '].v al ue _ c o unt s(), l abe l s=d f['Ge e r
'].v al ue _ c o unt s().i nd e x , auto pc t ='% 1.1f%
%',colors=['lightblue','lightcoral']) axes[0,1].set_title("Gender
Distribution")
#plot 3:Age distribution
a xe s[1, 0].hi st (d f['A ge '].d ro pna (), bi ns=8, c o l o r='l i ght gre e e d
ge c o l o r='bl ac k')
axes[1,0].set_title("Age Distribution") axes[1,0].set_xlabel("Age")
axes[1,0].set_ylabel("Frequency")
#plot 4:survival count
axe s[1, 1].bar(d f['Sur v i v e d '].v al ue _ c o unt s().i nd e x , d f['Su
ved'].value_counts(),color=['lightblue','lightcoral'])
axes[1,1].set_title("Survival Count")
axes[1,1].set_xlabel("Survived (0 = No, 1 = Yes)")
axes[1,1].set_ylabel("Count")
#plot 5:fare vs age
```

axe s[2, 0].sc at t e r(d f['A ge '], d f['Fare '], c o l o r='o range ', e d ge l o rs='bl ac k')

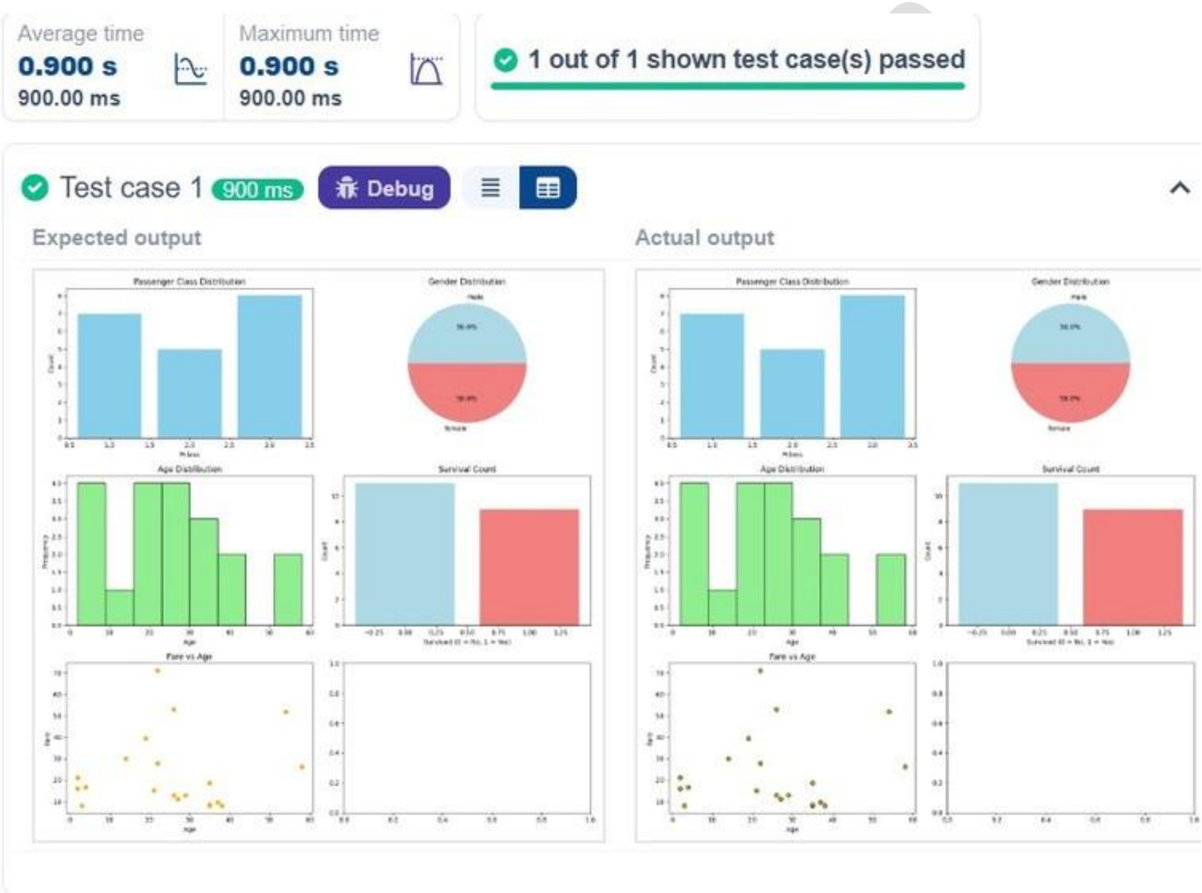
axes[2,0].set_title("Fare vs Age")

axes[2,0].set_xlabel("Age")

axes[2,0].set_ylabel("Fare")

plt.tight_layout() plt.show() O ut

put :

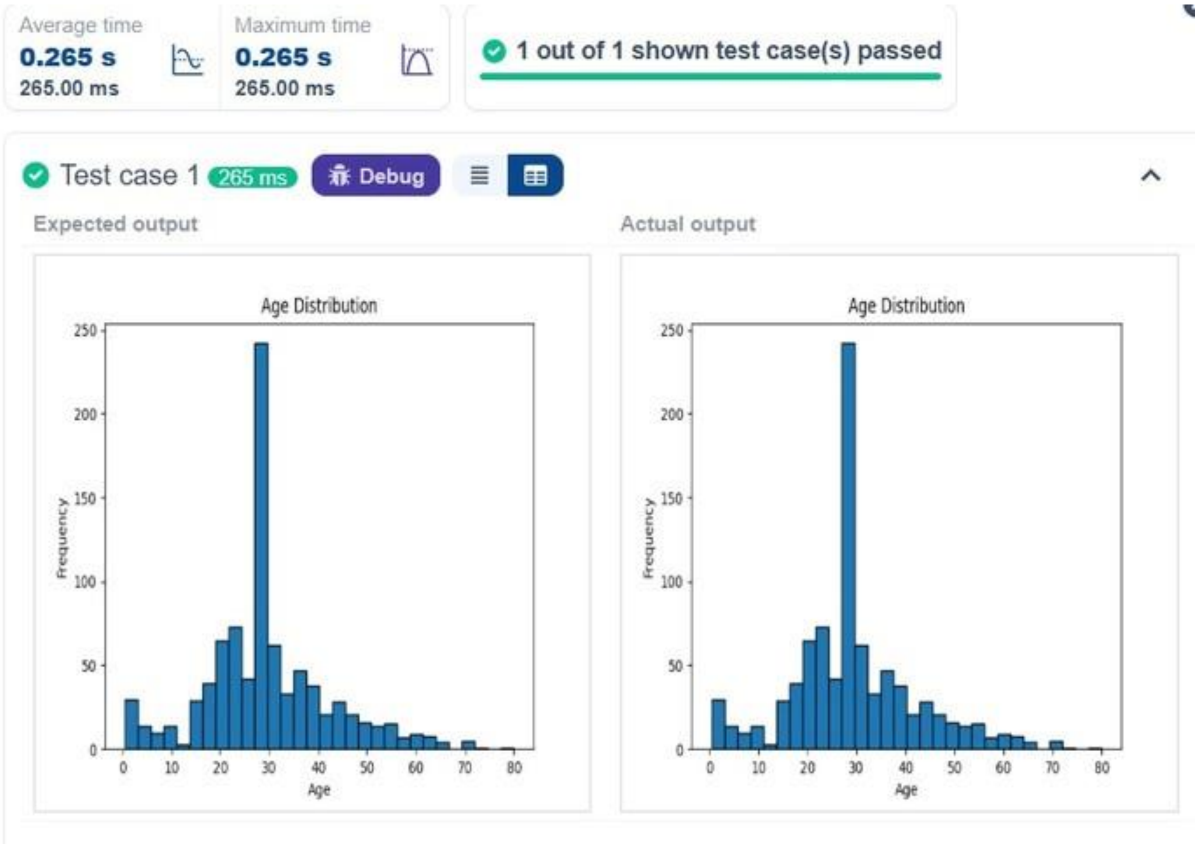


5.2.2 Histogram of Passenger information of Titanic

C od e:

```
#write your code here for histogram
import pandas as pd import
matplotlib.pyplot as plt

# Load the Titanic dataset data =
pd.read_csv('Titanic-Dataset.csv')
# Data Cleaning data['Age'].fillna(data['Age'].median(),inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True) # Convert categorical
features to numeric data['Sex'] = data['Sex'].map({'male': 0, 'female':
1}) data = pd.get_dummies(data, columns=['Embarked'], d ro p_ first
=True )
# Write your code here for
# Write your code here for Histogram #write
your code here for histogram
plt.hist(data['Age'],bins=30,edgecolor='k')
#plt.title('Age Distribution')
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.title('AgeDistribution')
plt.show() O ut put :
```



5.2.3 Bar plot of survival rate of passengers C

od e: import pandas as pd import

matplotlib.pyplot as plt

Load the Titanic dataset data =

pd.read_csv('Titanic-Dataset.csv')

Data Cleaning

data['Age'].fillna(data['Age'].median(),inplace=True)

data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True) data.drop('Cabin', axis=1, inplace=True) # Convert categorical features to numeric data['Sex'] = data['Sex'].map({'male': 0, 'female': 1}) data = pd.get_dummies(data, columns=['Embarked'], d ro p_ first =True)

Write your code here for Bar Plot for Survival Rate survival_counts = data['Survived'].value_counts() survival_counts.plot(kind='bar') plt.title('Survival Count') plt.xlabel('Survived') plt.ylabel('Count') plt.show() O ut put :

5.2.4 Bar plot for survival by Gender C od e:

```
import pandas as pd import
matplotlib.pyplot as plt # Load the Titanic
dataset data = pd.read_csv('Titanic-
Dataset.csv')
# Data Cleaning
data['Age'].fillna(data['Age'].median(),inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True) # Convert
categorical features to numeric data['Sex'] =
data['Sex'].map({'male': 0, 'female': 1}) data =
pd.get_dummies(data, columns=['Embarked'], d ro p_
first =True )
# Write your code here for Bar Plot for Survival by Ge
nd e r grouped=data.groupby('Sex')
['Sur v i v e d '].v al ue _ c o unt s().unst ac k()
grouped.plot (kind='bar', stacked=True)
plt.title('Survival by Gender') plt.xlabel('Gender')
plt.ylabel('Count')
plt.legend(['NotSurvived','Survived'])
plt.show() O ut put :
```

5.2.5 Bar plot for survival by Pclass C od e:

import pandas as pd import

matplotlib.pyplot as plt # Load the Titanic

dataset data = pd.read_csv('Titanic-

Dataset.csv')

Data Cleaning

data['Age'].fillna(data['Age'].median(),inplace=True)

data['Embarked'].fillna(data['Embarked'].mode()[0],

inplace=True) data.drop('Cabin', axis=1, inplace=True)

Convert categorical features to numeric data['Sex'] =

data['Sex'].map({'male': 0, 'female': 1})

data = pd.get_dummies(data, columns=['Embarked'], d ro p_ first =True)

Write your code here for Bar Plot for Survival by Pclass

survival_counts = data.groupby('Pclass')

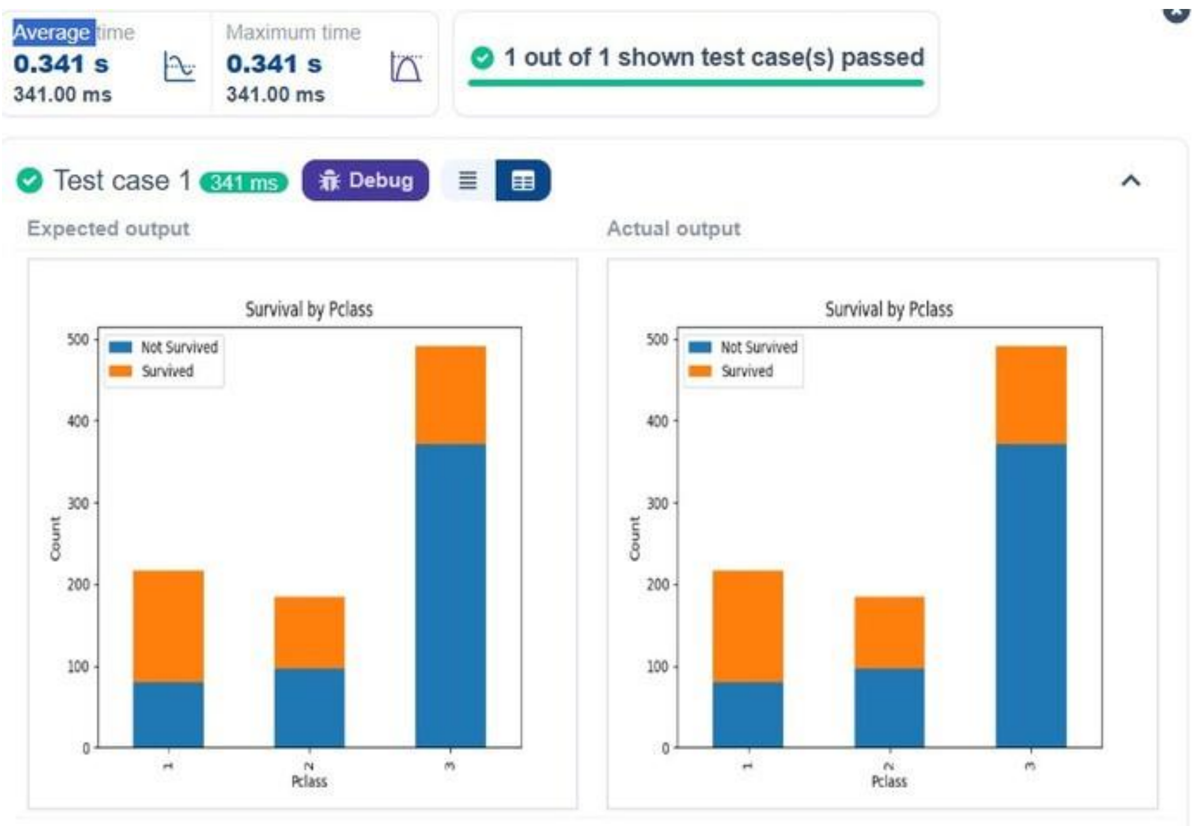
['Sur v i v e d '].v al ue _ c o unt s().unst ac k()

survival_counts.plot(kind='bar',stacked=True)

plt.title('Survival by Pclass') plt.ylabel('Count')

plt.legend(['Not Survived','Survived'])

plt.show() O ut put



5.2.6 Bar plot for survival by Embarked C od

e: import pandas as pd import

matplotlib.pyplot as plt # Load the Titanic

dataset data = pd.read_csv('Titanic-

Dataset.csv')

Data Cleaning

data['Age'].fillna(data['Age'].median(),inplace=True)

data['Embarked'].fillna(data['Embarked'].mode()[0],

inplace=True) data.drop('Cabin', axis=1, inplace=True)

```
# Convert categorical features to numeric data['Sex'] =

data['Sex'].map({'male': 0, 'female': 1})

data = pd.get_dummies(data, columns=['Embarked'], d ro
p_ first =True )

# Write your code here for Bar Plot for Survival by E

mbarke d survival_counts =

data.groupby('Embarked_Q')

['Sur v i v e d '].v al ue _ c o unt s().unst ac k()

survival_counts.plot(kind='bar',stacked=True)

plt.title('Survival by Embarked')

plt.xlabel('Embarked') plt.ylabel('Count')

plt.legend(['NotSurvived','Survived'])

plt.show() O ut put :
```

5.2.7 Box plot for Age Distribution C od e:

import pandas as pd import

matplotlib.pyplot as plt # Load the Titanic

dataset data = pd.read_csv('Titanic-

Dataset.csv')

Data Cleaning

data['Age'].fillna(data['Age'].median(),inplace=True)

data['Embarked'].fillna(data['Embarked'].mode()[0],

inplace=True) data.drop('Cabin', axis=1, inplace=True)

Convert categorical features to numeric data['Sex'] =

data['Sex'].map({'male': 0, 'female': 1})

data = pd.get_dummies(data, columns=['Embarked'], d ro
p_ first =True)

Write your code here for Box Plot for Age by Pclass

data.boxplot(column='Age',by='Pclass') plt.title('Age by

Pclass') plt.xlabel('Pclass') plt.ylabel('Age') plt.suptitle("")

plt.show() O ut put :

5.2.8 Box plot for Age by Survived C od e:

import pandas as pd import

matplotlib.pyplot as plt # Load the Titanic

dataset data = pd.read_csv('Titanic-

Dataset.csv')

Data Cleaning

data['Age'].fillna(data['Age'].median(),inplace=True)

data['Embarked'].fillna(data['Embarked'].mode()[0],

inplace=True) data.drop('Cabin', axis=1, inplace=True)

data = pd.get_dummies(data, columns=['Embarked'], d
ro p_ first =True)

Convert categorical features to numeric data['Sex'] =

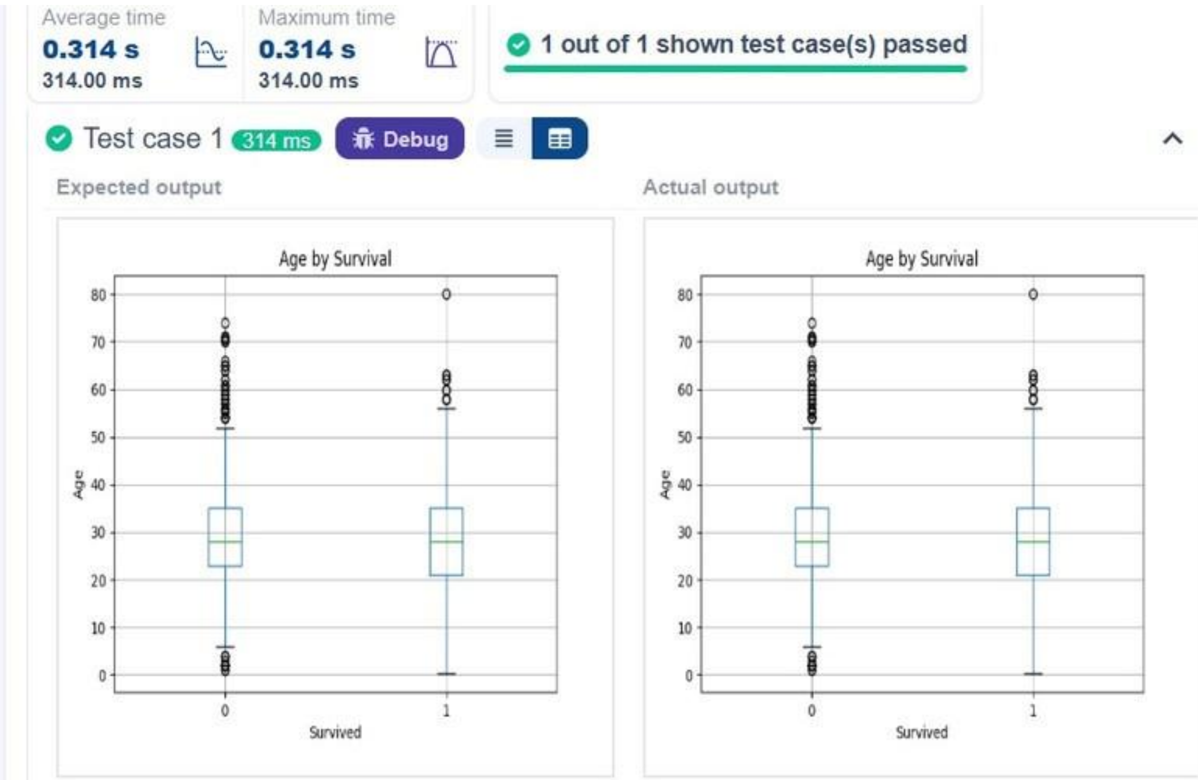
data['Sex'].map({'male': 0, 'female': 1})

Write your code here for Box Plot for Age by Survived d at

a.bo x pl o t (c o l umn='A ge ', by='Sur v i v e d ')

plt.title('Age by Survival') plt.xlabel('Survived')

plt.ylabel('Age') plt.suptitle("") plt.show() O ut put :



5.2.9 Box plot for fare by Pclass C od e:

import pandas as pd import

matplotlib.pyplot as plt # Load the Titanic

dataset data = pd.read_csv('Titanic-

Dataset.csv')

Data Cleaning

data['Age'].fillna(data['Age'].median(),inplace=True)

data['Embarked'].fillna(data['Embarked'].mode()[0],

inplace=True) data.drop('Cabin', axis=1, inplace=True) #

Convert categorical features to numeric data['Sex'] =

data['Sex'].map({'male': 0, 'female': 1}) data =

pd.get_dummies(data, columns=['Embarked'], d ro p_

first =True)

Write your code here for Box Plot for Fare by Pclass d at

a.bo x pl o t (c o l umn='Fare', by='Pc l ass') plt.title('Fare

by Pclass') plt.suptitle('') plt.xlabel('Pclass')

plt.ylabel('Fare')

plt.show()

O ut put :

5.2.10 Scatter plot for Age vs Fare C od e:

import pandas as pd import

matplotlib.pyplot as plt # Load the Titanic

dataset data = pd.read_csv('Titanic-

Dataset.csv')

Data Cleaning

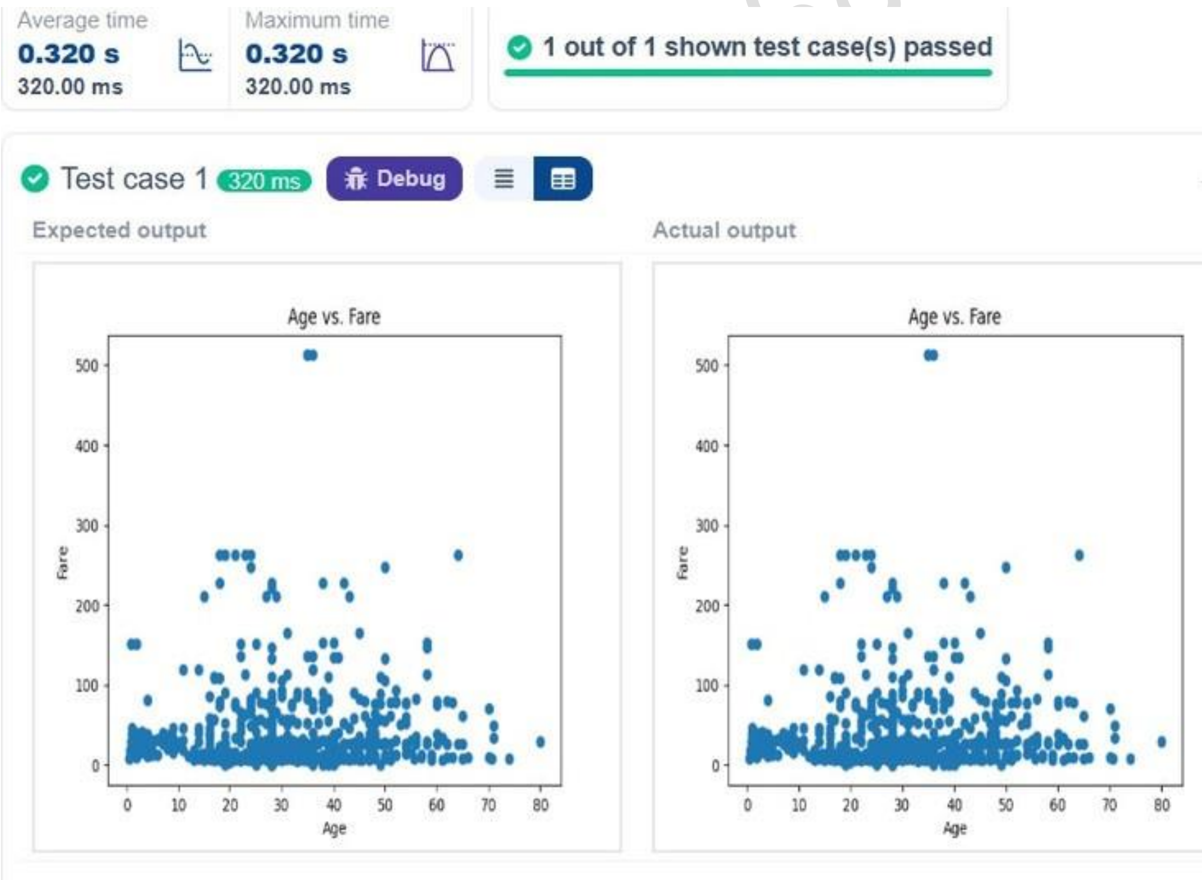
data['Age'].fillna(data['Age'].median(),inplace=True)

data['Embarked'].fillna(data['Embarked'].mode()[0],

inplace=True)

data.drop('Cabin', axis=1, inplace=True) # Convert categorical features to numeric data['Sex'] = data['Sex'].map({'male': 0, 'female': 1}) data = pd.get_dummies(data, columns=['Embarked'], d ro p_ first =True)

Write your code here for Box Plot for Fare by Pclass plt.scatter(data['Age'],data['Fare']) plt.title("Age vs. Fare") plt.xlabel("Age") plt.ylabel("Fare") plt.show() O ut put :



5.2.11 Scatter plot for Age vs Fare by Survived C od e: import pandas as pd import matplotlib.pyplot as plt # Load the Titanic dataset

data = pd.read_csv('Titanic-Dataset.csv')

Data Cleaning

data['Age'].fillna(data['Age'].median(),inplace=True)

data['Embarked'].fillna(data['Embarked'].mode()[0],

inplace=True) data.drop('Cabin', axis=1, inplace=True)

Convert categorical features to numeric data['Sex'] =

data['Sex'].map({'male': 0, 'female': 1}) data =

pd.get_dummies(data, columns=['Embarked'], d ro p_

first =True)

Write your code here for Scatter Plot for Age vs. Fare by

Survived colours=data['Survived'].map({0: 'red',1: 'blue'}) pl

t .sc at t e r(d at a['A ge '], d at a['Fare '], c =c o l o urs)

plt.title("Age vs. Fare by Survival") plt.xlabel("Age")

plt.ylabel('Fare') plt.show() O ut put :

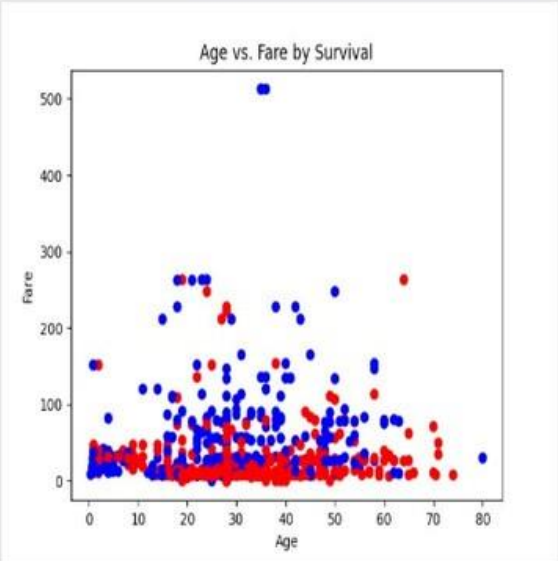
Average time
0.299 s
299.00 ms

Maximum time
0.299 s
299.00 ms

1 out of 1 shown test case(s) passed

Test case 1 299 ms Debug

Expected output



Actual output